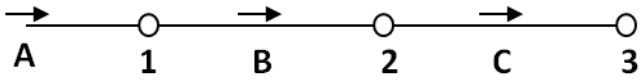
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Aeronautical Engineering**	**Course Type**	**C/FC/PE/OE**
Course Name	**Computational Fluid Dynamics**	**Course Code**	**AAE 4075**
Semester	**VII**	**Max Marks**	**50**
Date	**27/11/2024**	**Duration**	**3 Hours**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Explain the Neumann boundary condition with an example.	**2M**	1	1,2	1	1,2	2
1B	What are the local and convective derivatives? Explain their significance.	**3M**	1	1,2	1	1,2	2
1C	The inside and outside temperatures of 0.1 m thick metallic wall are 300°C and 35°C, respectively. Consider steady state condition, internal heat generation of 20,000 W/m^3 and thermal conductivity of wall material as 50 W/mK. Take spacing between the successive nodes as 0.025 m and determine the temperature distribution inside the metallic wall.	**5M**	3	1,2,3	3	1,2,3,13	5
2A	Write the continuity equation and explain its significance.	**2M**	1	1,2	1	1,2	3
2B	Explain the different steps involved while solving a problem using the CFD.	**3M**	2	1	2	1,2	3
2C	Consider the initial boundary value problem $$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$	**5M**	3	1,2,5	3	1,2,3,13	5

	Where, u is non-dimensional temperature, α is constant and t > 0. The initial conditions are $u(x,0) = e^{2x}$ for $0 < x < 1$ and boundary conditions are u(0, t) = 1, u(1, t) = 7.389 for all time, $t \geq 0$ Use, $r = \frac{\alpha t}{(\Delta x)^2} = 0.8 \; and \; \Delta x = 1/4$ Determine the non-dimensional temperatures using a suitable method.						
3A	Classify the partial differential equations based on their nature.	**2M**	2	1,2	2	1,2	2
3B	Explain the Alternating-Direction-Implicit method with a neat sketch.	**4M**	2	1,2	2	1,2	2
3C	Solve the following set of equations using the Gauss-Elimination method $2x + 2y + 4z = 18$ $x + 3y + 2z = 13$ $3x + y + 3z = 14$	**4M**	4	3,5	4	1,2,3	5
4A	Explain any one property of the control volume discretization scheme.	**2M**	2	1,2	2	1,2	2
4B	Obtain an expression for the face value of a control volume using the QUICK scheme.	**4M**	4	1,2	4	1,2	4
4C	In a composite wall, show that the interfact conductivity is taken as a harmonic mean of adjacent thermal conductivities while using the finite volume method.	**4M**	2	1,2,3	2	1,2	4
5A	When the fundamental governing equations are applied to different physical problems, they give different solutions. Explain the reason for the same.	**2M**	2	1,2	2	1,2	4
5B	Derive an expression for the velocity correction equation in the x direction.	**4M**	4	1,2,3	4	1,2	4
5C	In a steady, one dimensional, constant density situation as shown in below given figure, the velocity u is calculated for locations A, B and C, while the pressure p is calculated for locations 1, 2 and 3. The velocity correction formula is	**4M**	4	3,4,5	4	1,2,3,13	5

	$u = u^* + (p'_i - p'_{i+1})d$ Where, the locations i and i+1 lie on either side of the location for u. The value of d is 2 everywhere. The boundary conditions are $u_A = 10$ and $p'_3 = 0$. If, at a given stage in the iteration process, the momentum equations give $u^*_B = 8$ and $u^*_C = 11$, formulate the problem using SIMPLE Algorithm to obtain the linear equations at the grid points.  						

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***8***
2	***17***
3	***9***
4	***16***
5	-

BL	Total Marks
2	13
3	6
4	14
5	18